School of Computer & Systems Sciences JNU
MCA I Semester Mid-Sem Examination 2022-23

Time: 1 Hrs. Total Marks: 40

Note: Marks are indicated against each question.

Question No. 1 (8)

If P, Q and R are three atomic variables, obtain the Principal Disjunctive Normal form and Principal Conjunctive Normal form for the given formula without using TT.

$$(P \rightarrow (Q \wedge R)) \vee (\sim P \rightarrow (Q \vee R)$$

Question no. 2 (8)

Determine if the following inference pattern is Valid OR Invalid. Produce evidence (write the steps) to confirm its validity/invalidity.

$$\frac{\begin{array}{l}(p \vee q) \rightarrow (s \wedge t) \\ [\sim((\sim s) \vee (\sim t))] \rightarrow [(\sim r) \vee q]\end{array}}{(p \vee q) \rightarrow (r \rightarrow q)}$$

Question no. 3 (2+6)

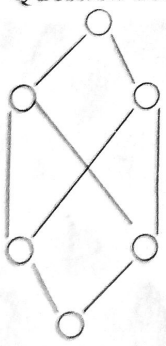
What is a partition on a set? Let A = {1,2,3,4,5} and a relation R defined on A is R = {(1,1), (1,2), (2,1), (2,2), (3,3), (4,4), (4,5), (5,4), (5,5). Obtain the cells of the partition.

Question no. 4 (1+3+4)

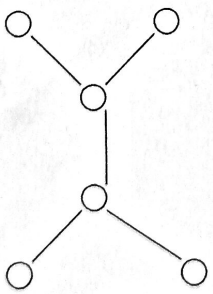
What is a Hasse diagram? Draw the Hasse diagrams of the following sets under the partial ordering relation "divides" and indicate those which are totally ordered. Also, determine least member, greatest member, minimal member and maximal member in each case.

(i) {2, 6, 24}
(ii) {1,2,3,6,12}
(iii) {3,9,27,54}

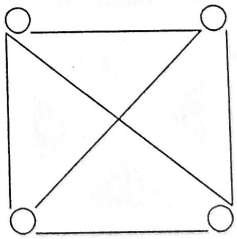
Question no. 5 What is a lattice? Determine (show steps), which of the following graphs are lattice? (2+6)



(a)



(b)



(c)

School of Computer & Systems Science
CS-404: Discrete Mathematics
Mid-Semester Exam

Date: 10/11/2025 Max Marks: 25
Time: 9:30 – 11:00am

ANSWER QUESTIONS OF **MAXIMUM MARKS 25.**

1) Give the set of logical constants. Give one each and describe the well-formed formula for propositional logic and predicate calculus. [3]

2) Let C be a subset of a universal set $U = \{u_1, u_2, u_3, \dots, u_n\}$. Name, describe and state application of the function f_A defined as [2]

$$f_A(u_i) = \begin{cases} 1 & if\ u_i \in A \\ 0 & if\ u_i \notin A \end{cases}$$

3) In a given counting problem, how can you decide the use of permutation or combination? A *Hackathon2026* must have **3 PhDs** and **5 MTech** students. Given **24 PhD** and **20 M.Tech.** eligible applicants, how many Hackathon2026 teams are possible? [3]

4) Let A = {a, b, c, d}, B = {1, 2, 3} and R_1 = {(a, 2), (b, 1), (c, 2), (d, 1)}. Is R_1 a Function? Is R_1^{-1} a function? Justify your answer. [2]

5) Given a Boolean alphabet B = {0, 1}, define a *regular expression* considering the set of standard operators – {∘, +,*}. What is the length of the shortest string represented by the (1∘ 0)*(1∘ 0 + 1*) (0*∘ 1), justifying your answer. [3+2]

6) Differentiate *Relation* from *Function*. Let A = $\mathbb{R}$ and let F = {(a, 2a+1)| ∀a a∈A}. What are the properties of F? Is F a relation or a function? Justify your answer. [5]

7) Let D = {1, 2, 3, 4, 5, 6} and for given cycles p_1 = (5, 1, 4) and p_2 = (2, 6, 3) complete the following and name the type of function these are. Give $p_1 \circ p_2$.

   a) (5, 1, 4) = $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ - & - & - & - & - & - \end{pmatrix}$

   b) (2, 6, 3) = $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ - & - & - & - & - & - \end{pmatrix}$ [5]

8) Consider the following and name the basic operators used to compute the set S and matrix $M_1 \odot M_2$. Give the total number of times the basic operators will be executed to compute S and $M_1 \odot M_2$.

i) Let U = {a, b, c, d, e, f, g, h, I, j}, G = {a, b, d, f}, H = {a, b, c, h. j}, I = {g, h}. To compute, $S = (\bar{G} \cap H) \cup I$.

ii) Let $M_1 = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ and $M_2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 \end{bmatrix}$. Let $\odot$ be the operator Boolean product A $\odot$ B for m×n Boolean matrices say A, m× p matrix and B, p×n matrix, be denoted by C = [c_{ij}] and defined by,

$$c_{ij} = \begin{cases} 1 & if\ a_{ik} = 1\ and\ b_{kj} = 1\ for\ some\ k, \quad 1 \le k \le p \\ 0 & otherwise \end{cases}$$

[3+5]

9) Let $R \subseteq Z \times Z$ defined by (a, b) R (c, d) if and only if a+d = b+c. Give the properties of R. Give three applications of directed graphs in Computer Science.

[2+3]

10) Differentiate the **transitive closure** of an arbitrary **set** S from an arbitrary **relation** R. Give the transitive closure of R_2 = {(1, 1), (1, 3), (2, 1), (2, 3), (2, 4), (3, 2), (3, 4) (4, 1)}.

[3]

11) Give full account of a **Hasse diagram**. Is *lattice* related to *Hasse* diagram? Justify your answer.

[5]

School of Computer and Systems Sciences
MCA – Semester–I (Monsoon 2026)
CS – 404: Discrete Mathematics

Mid-Semester Examination

Date: 15-10-2026 **Max Marks: 25**

Put down your Enrolment No. on the Answer sheet. No name.

Question 1 is compulsory. Answer **any FOUR of the remaining** questions.

Q1. Answer **any TEN** of the following. Each carry **1 mark**. [10]

a) Let **S** = {{a}, {b, c}}. Give the smallest set, say **A**, such that the **S** is a subset of the Powerset of A.

b) Consider the statement: The triple (i, j, d) represents the edge (i, j) with distance d. Represent the following set of triples in a way readable by a computer (that can be given as input to a program).

{(1, 2, 20), (1, 4, 5), (2, 3, 10), (3, 4, 15), (4, 3, 9), (4, 2, 32)}

c) Consider for yourself a set **A**, $|\mathbf{A}| = 16$. Define the characteristic function C. Represent **B** for $\mathbf{B} \subset \mathbf{A}$.

d) Given the set of all *uppercase letters, lowercase letters, digits*, and eight *special characters*. How many passwords of length nine can be generated?

e) Give a Boolean expression of four Boolean variables. What is the practical use of a Boolean expression in Conjunctive Normal Form?

f) Consider the following propositional variables – **P**: *Michelle wins at Olympics*; **Q**: *everyone admires Michelle*; **R**: *Michelle will get rich*; **S**: *Michelle's effort was in vain*. Express the following in an English sentence

$$((P \rightarrow (Q \wedge R)) \wedge ((\sim P) \rightarrow S))$$

g) Consider the following, where **j** and **p** are constants and **x** is a variable. Which type of formula in predicate calculus do the following i) – iii) belong? Name it. Further, use these to make one inference.

i) $\forall \mathbf{x}\,[F(\mathbf{j}, \mathbf{x}) \rightarrow (S(\mathbf{x}, \mathbf{j}) \vee D(\mathbf{x}, \mathbf{j}))]$

ii) $F(\mathbf{j}, \mathbf{p})$

iii) $\sim D(\mathbf{p}, \mathbf{j})$

h) Name a *rule of inference* that is similar to the *Direct Methods of proof*. State the two.

i) Define a *Poset* and give an example.

j) Let **T** be a set of all bit-strings of length 6. Consider a binary bit-string operator $\oplus$, using a bit-wise operator $\vee$ (or) (eg. $\mathbf{0} \vee \mathbf{1}$). For bit-strings $\mathbf{x}, \mathbf{y} \in \mathbf{T}$, let $\mathbf{x} \oplus \mathbf{y} \in \mathbf{T}$. Is $(\mathbf{T}, \oplus)$ a semigroup, monoid or a group? Justify.

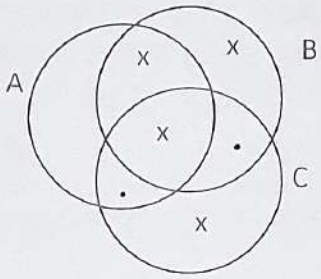
k) Give a working example (application) of the Pigeonhole Principle in computer science.

l) Give the basis on which Euclid devised a recursive method to compute the greatest common divisor of two natural numbers.

Q2. Answer the following, given the three sets A, B and C. [2+3]

a) Let the cardinality of the union of the three sets be 280. Let $|A| = 100$, $|B| = 200$ and $|C| = 150$. Further, let $|A \cap C| = 50$, $|A \cap C| = 80$ and $|B \cap C| = 90$. What is the cardinality of the intersection of A, B and C?

b) Give the expression for the region indicated by **x** in the following Venn diagram.



Q3. Answer the following, [2 +3]

a) Give an example of a relation that is *irreflexive* and *antisymmetric*. Prove the properties.

b) Consider *Odd* and *Even*, the set of odd and even natural numbers, respectively. Consider a function $f: Odd \rightarrow Even$ defined by $f(x) = x - 1$. Prove that f^{-1} exists. Define f^{-1}.

Q4. Which concept in Discrete Math is - *Tautology* or *Contradiction* or *Satisfiable* (*Contingent*)? Using a truth table determine which one of the above,

$\sim(P \vee (Q \wedge R)) \Leftrightarrow ((P \vee Q) \wedge (P \vee R))$ belongs to. [2+3]

Q5. State the *principle of Mathematical Induction* and define *well-formed formula* in predicate calculus. Represent the (statement of) principle of mathematical induction as a well-formed formula in predicate calculus. [2.5+2.5]

Q6. Let $(\mathbb{Z}, +)$ be an algebra. For $\mathbf{x}, \mathbf{y}, \mathbf{n} \in \mathbb{Z}$. [3+2]

a) Given $\mathbf{x}$ *mod* $\mathbf{n} = \mathbf{y}$ *mod* $\mathbf{n}$. Write down your observations. Based on your observations name the algebra $(\mathbb{Z}, +)$.

b) Let $\mathbf{a}$ *mod* $\mathbf{n} \in \mathbb{Z}$ and $\mathbf{b}$ *mod* $\mathbf{n} \in \mathbb{Z}$. What does $\mathbf{a} + \mathbf{b}$ imply?